

Constants

Permittivity of Free Space:

$$\varepsilon_0 = 8.854 \cdot 10^{-12} \frac{\text{As}}{\text{Vm}}$$

Permeability of Free Space:

$$\mu_0 = 4\pi \cdot 10^{-7} \frac{\text{Vs}}{\text{Am}}$$

Velocity of Light:

$$c = 2.998 \cdot 10^8 \frac{\text{m}}{\text{s}}$$

Boltzman Constant:

$$k = 1.381 \cdot 10^{-23} \frac{\text{J}}{\text{K}}$$

Wave Propagation on Lines

Propagation Constant

$$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')} = \alpha + j\beta$$

Line Impedance

$$Z_0 = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}}$$

Wave Length

$$\lambda = \frac{2\pi}{\beta}$$

Propagation Velocity

$$v = \frac{\omega}{\beta}$$

$$v = f \cdot \lambda$$

Velocity of TEM-Wave

$$v = \frac{1}{\sqrt{\varepsilon\mu}} = \frac{c}{\sqrt{\varepsilon_r}}$$

Reflection Coefficient

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

VSWR

$$VSWR = \frac{U_{max}}{U_{min}} = \frac{1+|\Gamma|}{1-|\Gamma|}$$

$$|\Gamma| = \frac{VSWR-1}{VSWR+1}$$

Input impedance of a terminated line

$$Z_{in} = Z_0 \frac{Z_L + Z_0 \tanh \gamma l}{Z_0 + Z_L \tanh \gamma l}$$

for lossless structures with $\gamma = j\beta$:

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$$

 $\lambda/4$ -Transformer

$$Z_{in} = \frac{Z_0^2}{Z_L}$$

Line with short

$$Z_{in} = Z_0 \tanh \gamma l = jZ_0 \tan \beta l$$

Line with open

$$Z_{in} = Z_0 \coth \gamma l = -jZ_0 \cot \beta l$$

Rectangular Waveguide Dominant TE₁₀ Mode:

Guided wave length:

$$\lambda_g = \frac{\lambda_0}{\sqrt{1 - \left(\frac{\lambda_0}{2a}\right)^2}}$$

Cut-Off wave length:

$$\lambda_c = 2a$$

Cut-Off frequency

$$f_c = \frac{c}{\lambda_c} = \frac{c}{2a}$$

Phase velocity

$$v_p = \frac{c}{\sqrt{1 - \left(\frac{\lambda_0}{2a}\right)^2}}$$

Group velocity

$$v_g = c\sqrt{1 - \left(\frac{\lambda_0}{2a}\right)^2}$$

Phase Constant

$$\beta = \frac{2\pi}{\lambda_g} = \frac{2\pi}{\lambda_0} \sqrt{1 - \left(\frac{\lambda_0}{2a}\right)^2}$$

Attenuation below cut-off (evanescent mode)

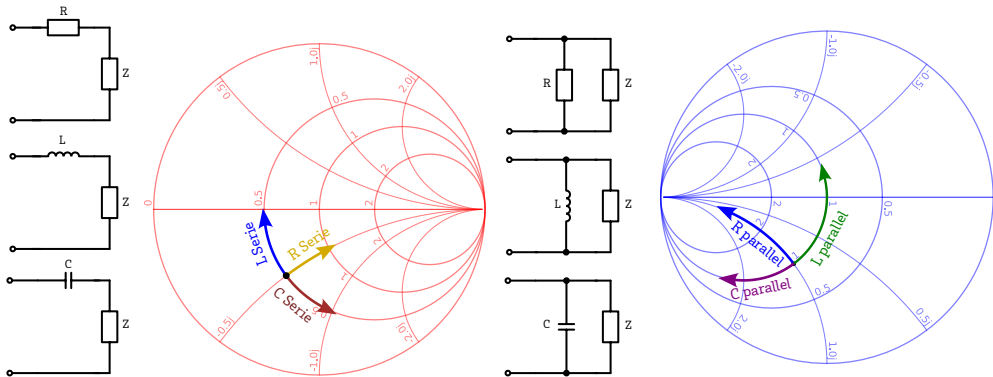
$$\alpha = \frac{2\pi}{\lambda_0} \sqrt{\left(\frac{\lambda_0}{2a}\right)^2 - 1}$$

Attenuation in prog. TE₁₀-mode

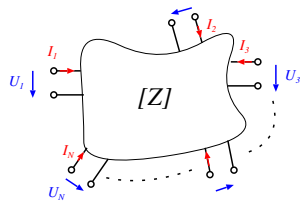
$$\alpha = \frac{R_0}{a^3 b \beta k \eta_0} (2b\pi^2 + a^3 k^2) [Np/m]$$

	TE_{mn} Mode	TM_{mn} Mode
k	$\omega\sqrt{\mu\epsilon}$	$\omega\sqrt{\mu\epsilon}$
k_c	$\sqrt{(m\pi/a)^2 + (n\pi/b)^2}$	$\sqrt{(m\pi/a)^2 + (n\pi/b)^2}$
β	$\sqrt{k^2 - k_c^2}$	$\sqrt{k^2 - k_c^2}$
λ_c	$\frac{2\pi}{k_c}$	$\frac{2\pi}{k_c}$
λ_g	$\frac{2\pi}{\beta}$	$\frac{2\pi}{\beta}$
v_p	$\frac{\omega}{\beta}$	$\frac{\omega}{\beta}$
α_d	$\frac{k^2 \tan \delta}{2\beta}$	$\frac{k^2 \tan \delta}{2\beta}$
E_z	0	$B_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$
H_z	$A_{mn} \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$	0
E_x	$\frac{j\omega\mu n\pi}{k_c^2 b} A_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$	$\frac{-j\beta m\pi}{k_c^2 a} B_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$
E_y	$\frac{-j\omega\mu m\pi}{k_c^2 a} A_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$	$\frac{-j\beta n\pi}{k_c^2 b} B_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$
H_x	$\frac{j\beta m\pi}{k_c^2 a} A_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$	$\frac{j\omega\epsilon n\pi}{k_c^2 b} B_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$
H_y	$\frac{j\beta n\pi}{k_c^2 b} A_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$	$\frac{-j\omega\epsilon m\pi}{k_c^2 a} B_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$
Z	$Z_{TE} = \frac{k\eta}{\beta}$	$Z_{TM} = \frac{\beta\eta}{k}$

Smith Chart



Network Analyses



$$\begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_N \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1N} \\ Z_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ Z_{N1} & \cdots & \cdots & Z_{NN} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} & \cdots & Y_{1N} \\ Y_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ Y_{N1} & \cdots & \cdots & Y_{NN} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_N \end{bmatrix}$$

Calculation of elements

$$Z_{ij} = \frac{V_i}{I_j} \Big|_{I_k=0 \text{ if } k \neq j}$$

$$Y_{ij} = \frac{I_i}{V_j} \Big|_{V_k=0 \text{ if } k \neq j}$$

$$\begin{bmatrix} V_1^- \\ V_2^- \\ \vdots \\ V_N^- \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1N} \\ S_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ S_{N1} & \cdots & \cdots & S_{NN} \end{bmatrix} \begin{bmatrix} V_1^+ \\ V_2^+ \\ \vdots \\ V_N^+ \end{bmatrix}$$

$$S_{ij} = \frac{V_i^-}{V_j^+} \Big|_{V_k^+=0 \text{ wenn } k \neq j}$$

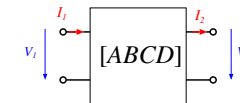
$$[S] = [S]^T \text{ for reciprocal Networks}$$

$$[S][S]^* = [U]$$

$$|S_{21}|^2 = 1 - |S_{11}|^2$$

for lossless networks

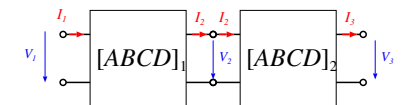
ABCD-Matrix



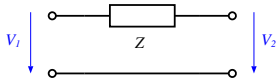
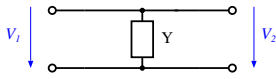
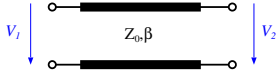
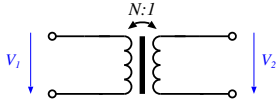
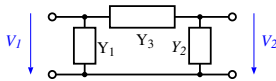
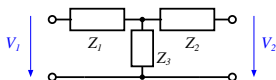
$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}$$

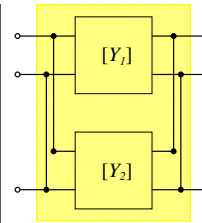
$$\det(ABCD) = 1$$

Cascading

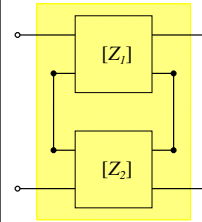


$$[ABCD]_{ges} = [ABCD]_1 \cdot [ABCD]_2 \cdots [ABCD]_N$$

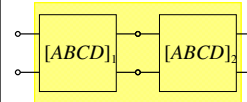
Circuit	ABCD-Parameter	
	$A = 1$ $C = 0$	$B = Z$ $D = 1$
	$A = 1$ $C = Y$	$B = 0$ $D = 1$
	$A = \cos \beta l$ $C = jY_0 \sin \beta l$	$B = jZ_0 \sin \beta l$ $D = \cos \beta l$
	$A = N$ $C = 0$	$B = 0$ $D = \frac{1}{N}$
	$A = 1 + \frac{Y_2}{Y_3}$ $C = Y_1 + Y_2 + \frac{Y_1 Y_2}{Y_3}$	$B = \frac{1}{Y_3}$ $D = 1 + \frac{Y_1}{Y_3}$
	$A = 1 + \frac{Z_1}{Z_3}$ $C = \frac{1}{Z_3}$	$B = Z_1 + Z_2 + \frac{Z_1 Z_2}{Z_3}$ $D = 1 + \frac{Z_2}{Z_3}$



$$[Y_{ges}] = [Y_1] + [Y_2] = \begin{bmatrix} Y_{1,11} + Y_{2,11} & Y_{1,12} + Y_{2,12} \\ Y_{1,21} + Y_{2,21} & Y_{1,22} + Y_{2,22} \end{bmatrix}$$



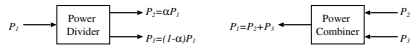
$$[Z_{ges}] = [Z_1] + [Z_2] = \begin{bmatrix} Z_{1,11} + Z_{2,11} & Z_{1,12} + Z_{2,12} \\ Z_{1,21} + Z_{2,21} & Z_{1,22} + Z_{2,22} \end{bmatrix}$$



$$[ABCD]_{ges} = [ABCD]_1 \cdot [ABCD]_2$$

$$= \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \cdot \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix}$$

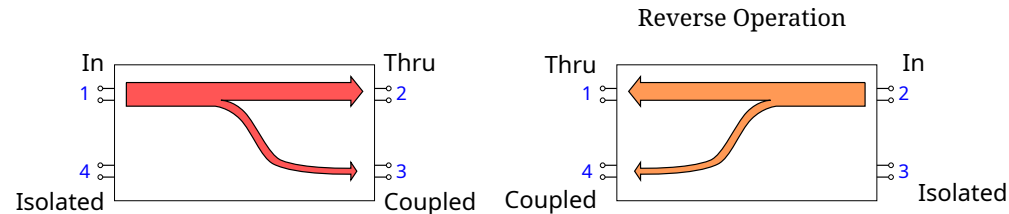
Symmetric Divider



S-Matrix

$$S = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Coupler



S-Matrix

$$S = \begin{bmatrix} 0 & \alpha & \beta & 0 \\ \alpha & 0 & 0 & -\beta \\ \beta & 0 & 0 & \alpha \\ 0 & -\beta & \alpha & 0 \end{bmatrix}$$

Coupling

$$C = 10 \log \frac{P_1}{P_3} = -20 \log \beta \text{ in dB}$$

Directivity

$$D = 10 \log \frac{P_3}{P_4} = 20 \log \frac{\beta}{|S_{14}|} \text{ in dB}$$

Isolation

$$I = 10 \log \frac{P_1}{P_4} = -20 \log |S_{14}| \text{ in dB}$$

Even-Odd-Analysis

Seperation of four-port in 2 two-ports
with even mode excitation

$$\rightarrow [S]_e$$

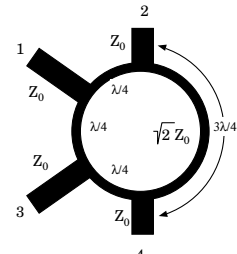
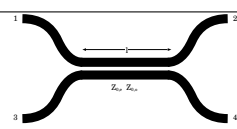
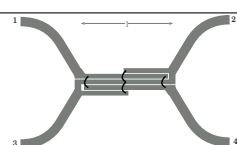
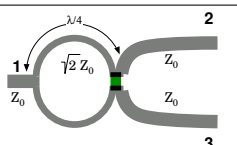
with odd mode excitation

$$\rightarrow [S]_o$$

total matrix

$$S_{11} = \frac{1}{2} (S_{11,e} + S_{11,o}) \quad S_{12} = \frac{1}{2} (S_{12,e} + S_{12,o})$$

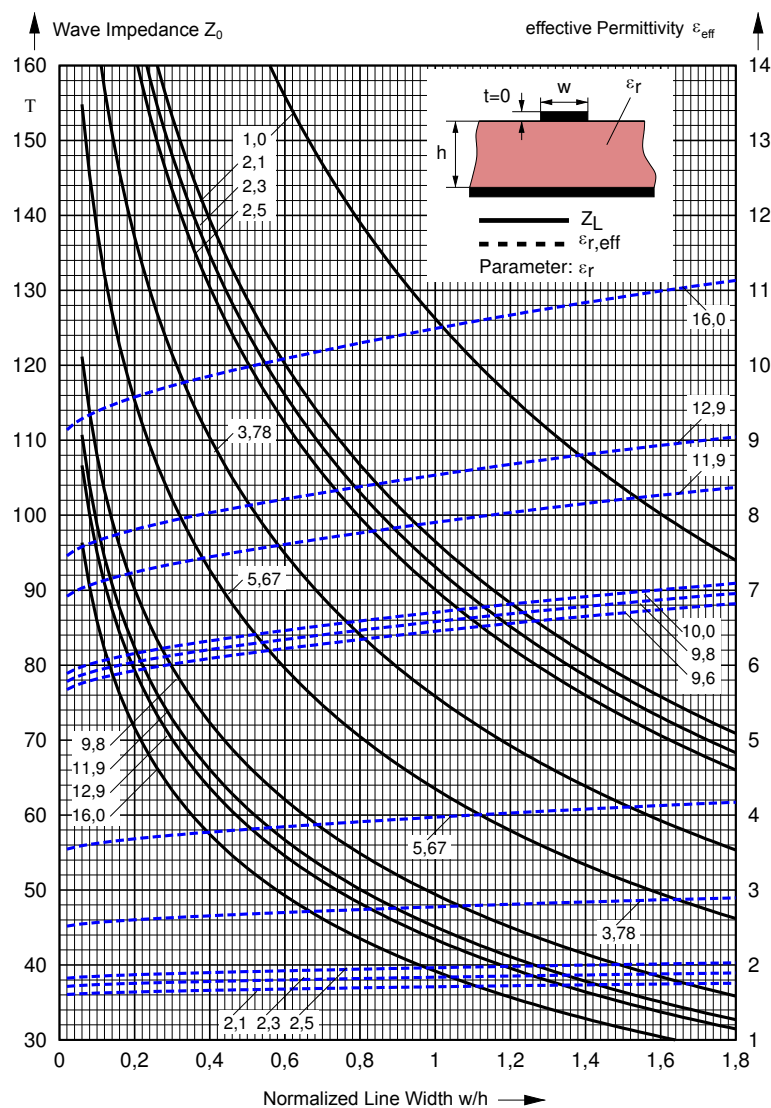
$$S_{13} = \frac{1}{2} (S_{11,e} - S_{11,o}) \quad S_{14} = \frac{1}{2} (S_{12,e} - S_{12,o})$$

90°-Hybrid		$S = \frac{-1}{\sqrt{2}} \begin{bmatrix} 0 & j & 1 & 0 \\ j & 0 & 0 & 1 \\ 1 & 0 & 0 & j \\ 0 & 1 & j & 0 \end{bmatrix}$
Rat Race/Mag-T		$[S] = -\frac{j}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{bmatrix}$
Line		$S = \begin{bmatrix} 0 & \alpha & j\beta & 0 \\ \alpha & 0 & 0 & j\beta \\ j\beta & 0 & 0 & \alpha \\ 0 & j\beta & \alpha & 0 \end{bmatrix}$
Lange		$S = \frac{-1}{\sqrt{2}} \begin{bmatrix} 0 & j & 1 & 0 \\ j & 0 & 0 & 1 \\ 1 & 0 & 0 & j \\ 0 & 1 & j & 0 \end{bmatrix}$
Wilkinson		$[S] = -\frac{j}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

	S	Z	Y	$ABCD$
S_{11}	S_{11}	$\frac{(Z_{11}-Z_0)(Z_{22}+Z_0)-Z_{12}Z_{21}}{\Delta Z}$	$\frac{(Y_0-Y_{11})(Y_0+Y_{22})+Y_{12}Y_{21}}{\Delta Y}$	$\frac{A+\frac{B}{Z_0}-CZ_0-D}{A+\frac{B}{Z_0}+CZ_0+D}$
S_{12}	S_{12}	$\frac{2Z_{12}Z_0}{\Delta Z}$	$\frac{-2Y_{12}Y_0}{\Delta Y}$	$\frac{2(AD-BC)}{A+\frac{B}{Z_0}+CZ_0+D}$
S_{21}	S_{21}	$\frac{2Z_{21}Z_0}{\Delta Z}$	$\frac{-2Y_{21}Y_0}{\Delta Y}$	$\frac{2}{A+\frac{B}{Z_0}+CZ_0+D}$
S_{22}	S_{22}	$\frac{(Z_{11}+Z_0)(Z_{22}-Z_0)-Z_{12}Z_{21}}{\Delta Z}$	$\frac{(Y_0+Y_{11})(Y_0-Y_{22})+Y_{12}Y_{21}}{\Delta Y}$	$\frac{-A+\frac{B}{Z_0}-CZ_0+D}{A+\frac{B}{Z_0}+CZ_0+D}$
Z_{11}	$Z_0 \frac{(1+S_{11})(1-S_{22})+S_{12}S_{21}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{11}	$\frac{Y_{22}}{ Y }$	$\frac{A}{C}$
Z_{12}	$Z_0 \frac{2S_{12}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{12}	$\frac{-Y_{12}}{ Y }$	$\frac{AD-BC}{C}$
Z_{21}	$Z_0 \frac{2S_{21}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{21}	$\frac{-Y_{21}}{ Y }$	$\frac{1}{C}$
Z_{22}	$Z_0 \frac{(1-S_{11})(1+S_{22})+S_{12}S_{21}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{22}	$\frac{Y_{11}}{ Y }$	$\frac{D}{C}$
Y_{11}	$Y_0 \frac{(1-S_{11})(1+S_{22})+S_{12}S_{21}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{Z_{22}}{ Z }$	Y_{11}	$\frac{D}{B}$
Y_{12}	$Y_0 \frac{-2S_{12}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{-Z_{12}}{ Z }$	Y_{12}	$\frac{BC-AD}{B}$
Y_{21}	$Y_0 \frac{-2S_{21}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{-Z_{21}}{ Z }$	Y_{21}	$\frac{-1}{B}$
Y_{22}	$Y_0 \frac{(1+S_{11})(1-S_{22})+S_{12}S_{21}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{Z_{11}}{ Z }$	Y_{22}	$\frac{A}{B}$
A	$\frac{(1+S_{11})(1-S_{22})+S_{12}S_{21}}{2S_{21}}$	$\frac{Z_{11}}{Z_{21}}$	$\frac{-Y_{22}}{Y_{21}}$	A
B	$Z_0 \frac{(1+S_{11})(1+S_{22})-S_{12}S_{21}}{2S_{21}}$	$\frac{ Z }{Z_{21}}$	$\frac{-1}{Y_{21}}$	B
C	$\frac{1}{Z_0} \frac{(1-S_{11})(1-S_{22})-S_{12}S_{21}}{2S_{21}}$	$\frac{1}{Z_{21}}$	$\frac{- Y }{Y_{21}}$	C
D	$\frac{(1-S_{11})(1+S_{22})+S_{12}S_{21}}{2S_{21}}$	$\frac{Z_{22}}{Z_{21}}$	$\frac{-Y_{11}}{Y_{21}}$	D
$ Z = Z_{11}Z_{22} - Z_{12}Z_{21} \quad Y = Y_{11}Y_{22} - Y_{12}Y_{21} \quad Y_0 = \frac{1}{Z_0} \quad \Delta = (Y_{11} + Y_0)(Y_{22} + Y_0) - Y_{12}Y_{21} \quad \Delta = (Z_{11} + Z_0)(Z_{22} + Z_0) - Z_{12}Z_{21}$				

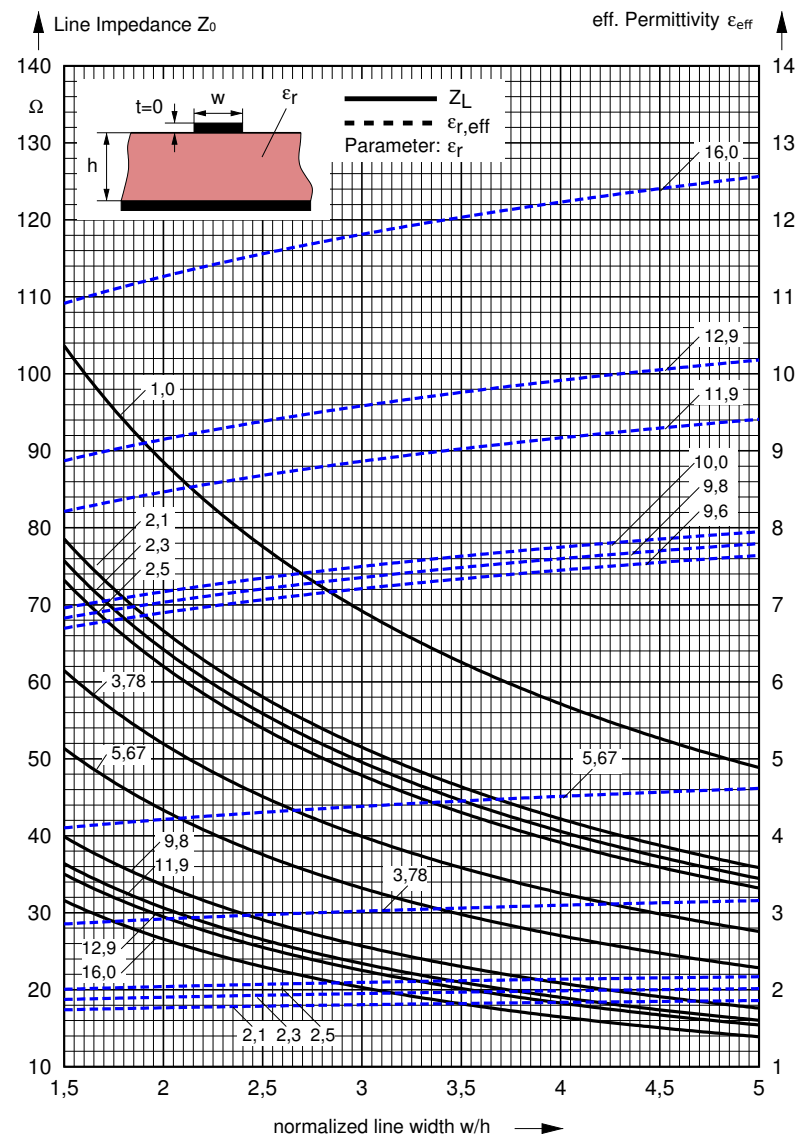
Tabelle 1: Twoport Parameter

Microstrip Line



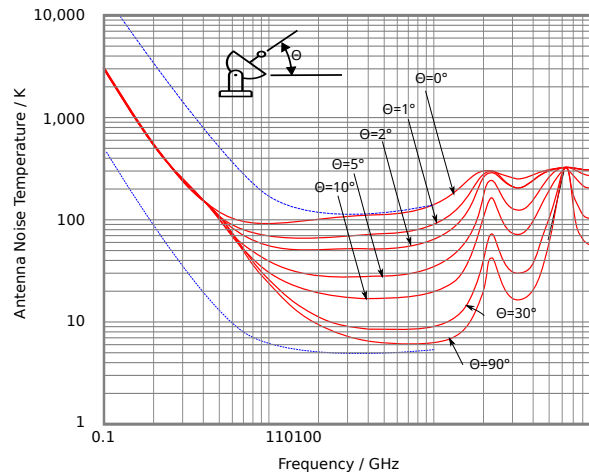
MS-line wave length

$$\lambda_{MS} = \frac{\lambda_0}{\sqrt{\epsilon_{eff}}} \quad v = \frac{c}{\sqrt{\epsilon_{eff}}}$$



Thermal Noise

Antenna Noise Temperature



Noise temperature at ground

$$T_A = 290\text{K}$$

Noise Figure

$$F = \frac{S_i/N_i}{S_o/N_o} \geq 1$$

with the condition input noise $T_a = 290\text{ K}$

Noise Temperature

$$T_e = (F - 1)T_0$$

Noise Temperature of passive

$$F = 1 + (L - 1)\frac{T}{T_0}$$

at room temperature:

$$F = L$$

Cascaded Two-Ports

$$F_{tot} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots$$

using noise temperature

$$T_{e,tot} = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 G_2} + \dots$$

Gain

$$G_{tot} = G_1 \cdot G_2 \cdot G_3$$

G over T

$$G/T = \frac{\text{Antenna Gain}}{\text{System Noise Temp}} = \frac{G_{ant}}{T_a + T_e}$$

$$SNR = \frac{S_0}{N_0} = \frac{EIRP \cdot \frac{1}{L_p} \cdot G_{ant} G_{rec}}{k (T_a + T_e) B G_{rec}} = EIRP \frac{1}{L_p} \frac{1}{k} \frac{1}{B} \frac{G}{T}$$

$$SNR_{dB} = EIRP_{dB} - L_{p,dB} - \underbrace{-10 \log(k) - 10 \log(B)}_{+228.6 \text{ K}} + \left. \frac{G}{T} \right|_{dB}$$

Amplifier Design

Power Gain

$$G = \frac{P_l}{P_{in}} = \frac{|S_{21}|^2 (1 - |\Gamma_l|^2)}{|1 - S_{22} \Gamma_l|^2 (1 - |\Gamma_{in}|^2)}$$

Available Power Gain

$$G_A = \frac{P_{av n}}{P_{av s}} = \frac{|S_{21}|^2 (1 - |\Gamma_s|^2)}{|1 - S_{11} \Gamma_s|^2 (1 - |\Gamma_{out}|^2)}$$

Transducer Power Gain

$$G_T = \frac{P_l}{P_{av s}} = \frac{|S_{21}|^2 (1 - |\Gamma_s|^2) (1 - |\Gamma_l|^2)}{|1 - S_{22} \Gamma_l|^2 |1 - \Gamma_s \Gamma_{in}|^2}$$

where

$$\Gamma_{in} = S_{11} + \frac{S_{12} S_{21} \Gamma_L}{1 - S_{22} \Gamma_L}, \quad \Gamma_{out} = S_{22} + \frac{S_{12} S_{21} \Gamma_S}{1 - S_{11} \Gamma_S}$$

Unilateral Case

$$G_{TU} = G_S G_0 G_L$$

where

$$G_S = \frac{1 - |\Gamma_S|^2}{|1 - S_{11} \Gamma_S|^2} \quad G_0 = |S_{21}|^2 \quad G_L = \frac{1 - |\Gamma_L|^2}{|1 - S_{22} \Gamma_L|^2}$$

Determinant of $[S]$

$$\Delta = S_{11} S_{22} - S_{12} S_{21}$$

Output Stability Circles

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}, \quad R_L = \left| \frac{S_{12} S_{21}}{|S_{22}|^2 - |\Delta|^2} \right|$$

Input Stability Circles

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}, \quad R_S = \left| \frac{S_{12} S_{21}}{|S_{11}|^2 - |\Delta|^2} \right|$$

K- Δ -Test:

$$\text{Stable if } K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12} S_{21}|} > 1, \text{ and } |\Delta| < 1.$$

μ -Test

$$\text{uncond. Stable if } \mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12} S_{21}|} > 1$$

Conjugate Matching $\Gamma_S = \Gamma_{in}^*$ and $\Gamma_L = \Gamma_{out}^*$

$$\Gamma_S = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}$$

where

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2, \quad B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*, \quad C_2 = S_{22} - \Delta S_{11}^*$$

Const. Gain Circles for Input

$$C_S = \frac{g_S S_{11}^*}{1 - (1 - g_S) |S_{11}|^2}, R_S = \frac{\sqrt{1 - g_S} (1 - |S_{11}|^2)}{1 - (1 - g_S) |S_{11}|^2}$$

Const. Gain Circles for Output

$$C_L = \frac{g_L S_{22}^*}{1 - (1 - g_L) |S_{22}|^2}, R_L = \frac{\sqrt{1 - g_L} (1 - |S_{22}|^2)}{1 - (1 - g_L) |S_{22}|^2}$$

Noise Figure

$$F = F_{min} + \frac{4R_N}{Z_0} \frac{|\Gamma_S - \Gamma_{opt}|^2}{(1 - |\Gamma_S|^2) |1 + \Gamma_{opt}|^2}$$

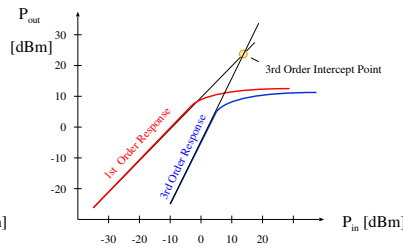
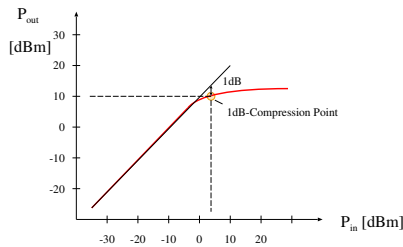
Const. Noise Figure Circles

$$C_F = \frac{\Gamma_{opt}}{N+1}, R_F = \frac{\sqrt{N(N+1 - |\Gamma_{opt}|^2)}}{N+1}$$

with

$$N = \frac{|\Gamma_S - \Gamma_{opt}|^2}{1 - |\Gamma_S|^2} = \frac{F - F_{min}}{4R_N/Z_0} |1 + \Gamma_{opt}|^2$$

Amplifier Dynamic Range and 3rd Order Intercept



output signal Non-lin

$$v_o(t) = a_0 + a_1 V_0 \cos \omega_0 t + a_2 V_0^2 \cos^2 \omega_0 t + a_3 V_0^3 \cos^3 \omega_0 t + \dots \quad (1)$$

$$= (a_0 + \frac{1}{2} a_2 V_0^2) + (a_1 V_0 + \frac{3}{4} a_3 V_0^3) \cos \omega_0 t + \frac{1}{2} a_2 V_0^2 \cos 2\omega_0 t \quad (2)$$

$$+ \frac{1}{4} a_3 V_0^3 \cos 3\omega_0 t + \dots \quad (3)$$

Voltage gain at signal frequency ω_0

$$G_v = \frac{v_o(\omega_0)}{v_{in}(\omega_0)} = \frac{a_1 V_0 + \frac{3}{4} a_3 V_0^3}{V_0} = a_1 + \frac{3}{4} a_3 V_0^2$$

Output power IP3 Point

$$P_3 = P_{\omega_1} |_{V_0=V_{IP3}} = \frac{1}{2} a_1^2 V_{IP}^2 = \frac{2|a_1^3|}{3|a_3|}$$

Antennas

Isotropic Antenna

Power flux density r

$$S_0(r) = \frac{P_s}{4\pi r^2} \text{ oder } \vec{S}(r) = \vec{E} \times \vec{H} = \frac{|E|^2}{\eta_0} \frac{\vec{r}}{|r|}$$

Antenna gain

$$G(\phi, \theta) = \frac{S(r, \phi, \theta)}{S_0(r)}$$

in Decibel

$$G_{dB} = 10 \log G$$

Directivity

$$D = G/\eta$$

$D = G$ für verlustlose Antennen

Krauss' Equation

$$D = \frac{4\pi}{\Theta_{BW} \Phi_{BW}}$$

Effective Antenna area

$$A_e = \frac{P_e}{S_e}$$

Effective Antennenarea using Gain

$$A_e = \frac{G_{max} \lambda^2}{4\pi}$$

Antenna Efficiency

$$\eta_{Ant} = \frac{A_{Phy}}{A_e}$$

Available Power

$$P_r = S_r A_r = S_r G \frac{\lambda^2}{4\pi}$$

Wireless Link

Transmit Power P_t generates power flux density

$$S_e = \frac{P_t G_t}{4\pi r^2}$$

Friis' Formula:

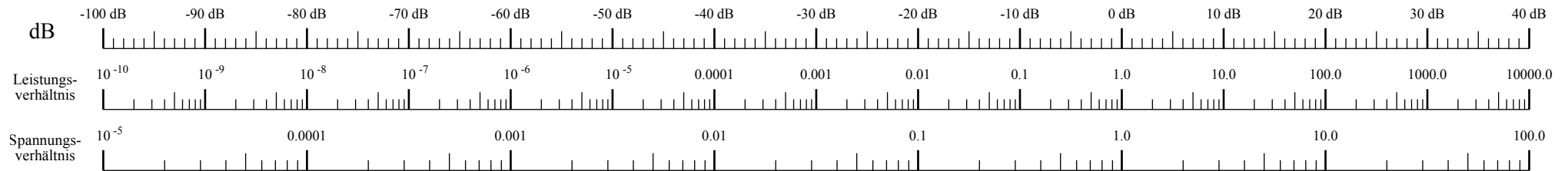
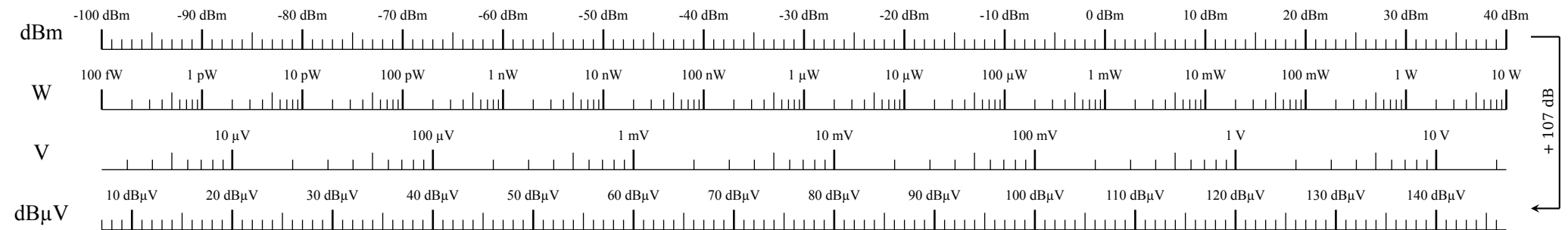
$$P_r = P_t G_t G_r \frac{\lambda^2}{(4\pi r)^2}$$

Prefixes and Greek Letters

Prefix	Symbol	Factor
Yocto	y	10^{-24}
Zepto	z	10^{-21}
Atto	a	10^{-18}
Femto	f	10^{-15}
Pico	p	10^{-12}
Nano	n	10^{-9}
Micro	μ	10^{-6}
Milli	m	10^{-3}
Centi	c	10^{-2}
Deci	d	10^{-1}
Deca	da	10^1
Hecto	h	10^2
Kilo	k	10^3
Mega	M	10^6
Giga	G	10^9
Tera	T	10^{12}
Peta	P	10^{15}
Exa	E	10^{18}
Zetta	Z	10^{21}
Yotta	Y	10^{24}

Uppercase	Name	Lowercase
A	Alpha	α
B	Beta	β
Γ	Gamma	γ
Δ	Delta	δ
E	Epsilon	ϵ
Z	Zeta	ζ
H	Eta	η
Θ	Theta	θ
I	Iota	ι
K	Kappa	κ
Λ	Lambda	λ
M	Mu	μ
N	Nu	ν
Ξ	Xi	ξ
O	Omicron	o
Π	Pi	π
P	Rho	ρ
Σ	Sigma	σ
T	Tau	τ
Υ	Upsilon	υ
Φ	Phi	ϕ
X	Chi	χ
Ψ	Psi	ψ
Ω	Omega	ω

Type	Image	Description	Frequency Range
BNC		Bayonet Neill–Concelman connector, commonly used in RF applications, test equipment, and video signals.	0-4 GHz
N-Type		A rugged RF connector, often used in higher frequency and power applications.	0-11 GHz
SMA		SubMiniature version A, used for microwave frequencies, widely used in wireless communication.	0-18 GHz
SMB		SubMiniature version B, used in applications requiring quick and easy connect/disconnect.	0-4 GHz
TNC		Threaded Neill–Concelman connector, similar to BNC but with a threaded coupling.	0-11 GHz
UHF		Ultra High Frequency connector, an older type of connector for RF signals.	0-300 MHz
MCX		Micro coaxial connector, used in applications with limited space.	0-6 GHz


Leistung und Spannung an 50 Ω Last:


Verhältnis	dB
1	0 dB
2	3 dB
3	4.77 dB
4	6 dB = 3+3
8	9 dB
$\frac{5}{4}$	1 dB
$\frac{1}{2}$	-3 dB
$\frac{1}{4}$	-6 dB

Verhältnis	dB
10	10 dB
100	20 dB
1000	30 dB
10^9	90 dB
$\frac{1}{10}$	-10 dB
$\frac{1}{100}$	-20 dB
$\frac{1}{1000}$	-30 dB
10^{-9}	-90 dB

Absolut	dBm	dBW	dBuV
1 mW	0 dBm	-30 dBW	107 dBuV
2 mW	3 dBm	-27 dBW	110 dBuV
10 mW	10 dBm	-20 dBW	117 dBuV
100 mW	20 dBm	-10 dBW	127 dBuV
1 W	30 dBm	0 dBW	137 dBuV
1 uW	-30 dBm	-60 dBW	77 dBuV
1 nW	-60 dBm	-90 dBW	47 dBuV
1pW	-90 dBm	-120 dBW	17 dBuV

Leistungsverhältnisse:

$$G_{dB} = 10 \log G \quad G = 10^{\frac{G_{dB}}{10}}$$

Spannungsverhältnisse:

$$G_U = \frac{U_2}{U_1} = \sqrt{G}$$

$$G_{U,dB} = 20 \log G_u \quad G_U = 10^{\frac{G_{U,dB}}{20}}$$

Pegel:

$$P_{dBW} = 10 \log \frac{P}{1W} \quad P = 10^{\frac{P_{dBW}}{10}} \cdot 1W$$

$$P_{dBm} = 10 \log \frac{P}{1mW} \quad P = 10^{\frac{P_{dBm}}{10}} \cdot 1mW$$

$$U_{dBuV} = 20 \log \frac{U}{1\mu V} \quad U = 10^{\frac{U_{dBuV}}{20}} \cdot 1\mu V$$

für Spannungsverhältnisse dB $\times 2$ nehmen